Colin Balfour

colinbalfour@gmail.com | colinbalfour.github.io | github.com/ColinBalfour

Education

Worcester Polytechnic Institute – MS in Robotics Engineering (4.0/4.0) May 2027
Worcester Polytechnic Institute – BS in Robotics Engineering, Minor in Mathematics (3.81/4.0) December 2026

Experience

Robotics Software Intern - IsaacSim, NVIDIA – Santa Clara, CA May 2026 – Present

- Designed refactor enabling generic Isaac Lab policy deployment for locomotion and manipulation in IsaacSim.
- Achieved **closed-loop parity for Spot, H1, Go2, and ANYmal-C** in simulation deployment by tracing subtle discrepancies across actuator dynamics, policy inference, solver behavior, and simulation configuration.
- Developed systematic A/B benchmarks of physics solver and simulation parameters, significantly improving Franka cabinet performance and driving cross-team fixes with Isaac Lab and Newton Physics engineers.

Perception and Deep Learning Researcher, PeAR Group (WPI) – Worcester, MA May 2024 – Present

- Researching **RL for drone navigation at up to 30m/s** with onboard online adaptation to new environments
- Recruited and led a 4 engineer infrastructure team supporting **15+ researchers**; delivered HPC integration, NFS-based shared storage, CI/CD pipelines, Dockerized workflows, and reproducible simulation environments.
- Built a **photorealistic** 3DGS-based drone simulator with seamless integration into NVIDIA Isaac Sim, Isaac Lab, and Blender, enabling high-fidelity sim-to-real policy training.
- Developed the **first** palm-sized drone capable of autonomous **navigation in complete darkness, smoke, and snow** using onboard ultrasound sensing and compute achieving **84% success** across 10 scenes (200+ trials)
- Introduced an onboard active vision-based navigation approach for quadcopters, with **80%** success in cluttered, unseen forest environments, enabling **zero-shot transfer sim2real** using hierarchical reinforcement learning
- Developed a **custom CUDA kernel** for rendering high quality depth images and collision checks in **<1ms**
- Optimized custom vision models on Google Coral TPU and Jetson Orin Nano for up to **3× faster performance**

Machine Learning Intern - ADAS Perception, Magna Electronics – Lowell, MA Jan 2026 – May 2026

- Optimized 520GB (VRAM) VLM-based synthetic data generation pipeline to run across 8× A100 (40GB) GPU cluster for **high-fidelity multi-view AV dataset generation and augmentation**, with containerized workflows
- Analyzed and created pipeline for comprehensive ablations on NVIDIA COSMOS (diffusion VLM) inputs: for different prompts, resolutions, # views, camera parameters, and various conditional inputs.
- Presented end-to-end pipeline for dataset creation to **30+** technical staff, with estimated **$3M/year** savings

Research Intern - Physical Sciences & Systems, RTX BBN Technologies – Cambridge, MA May 2025 – Dec 2025

- Prototyped and tested detection system for small targets at up to **60m** using TI automotive radar
- Built a robust deployment and monitoring system for layperson for a 4-server DAQ behind a bastion, with offline-safe auto-recovery and encrypted alerting, **reducing >16 hr/week of manual maintenance.**

Publications

- M. Velmurugan, P. Brush, **Colin Balfour**, R. Pryzbala, N. Sanket | ***Saranga: milliWatt Ultrasound for Navigation in Visually Degraded Environments on Palm-Sized Aerial Robots***, Science Robotics March 2026
- *[Under Review]* **Colin Balfour***, D. Singh*, N. Sanket | ***AttentionSeeker: Using Defocus in Events for Passive Attention Based Aerial Navigation***, IEEE Robotics and Automation Letters (RA-L) 2026
- *[Under Review]* **Colin Balfour**, K. Srivastava, D. Singh, N. Sanket | ***ActiveNav: Learning Active Monocular Flight in Forests***, IEEE Conference on Robotics and Automation (ICRA) 2027

Projects

Einstein Vision: 3D-Rendered Full Self-Driving Perception Pipeline github.com/ColinBalfour/Einstein-Vision

- Perception stack with rendered viz for a self-driving car, using a variety of models with only a single camera

Drone VLA: Language-based Navigation and Tracking at High Speeds (private, coming soon)

- Fine-tuned COSMOS Reason 256M using SFT then RL with asynchronous perception for low-latency navigation